THE COLLISION PERIODIC TABLE

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Abstract. For a prime $p > b^2$ not dividing the base b, the lag-1 signed collision count

$$S(p) = C(b \bmod p) - \left\lfloor \frac{p-1}{b} \right\rfloor$$

is an integer determined by $p \bmod b^2$. The $\varphi(b^2)$ values form the *collision periodic table*, a fixed invariant of the base.

The table satisfies the antisymmetry law $S(a) + S(b^2 - a) = -1$ for every coprime class a, the range is $[-(b-1),\ b-2]$, the mean is exactly $-1/2$, and exactly half the classes are negative. These properties are proved by elementary complement counting and floor arithmetic. A brief final note records the later analytic interpretation developed in 2026.

1. Introduction

This paper exhibits the lag-1 collision periodic table: a finite, computable table that assigns an integer to each coprime residue class modulo b^2 and determines the collision behavior of every sufficiently large prime.

The point is elementary. At lag 1, the specific collision count at multiplier b can be written as a finite slice count depending only on the residue class of p modulo b^2. This produces a fixed table with $\varphi(b^2)$ cells, one integer per coprime class.

The resulting table is not approximate. It is exact: an integer per class, constant for all primes in that class, with proved structural properties. In base 10, the table has 40 entries indexed by the coprime classes modulo 100. The last two decimal digits of a prime determine its collision fingerprint.

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2. The Fingerprint Theorem

Theorem 1 (Finite determination at lag 1)**.** *For any base $b \geq 2$ and prime $p > b^2$ with $p \nmid b$, the signed collision count*

$$S(p) = C(b \bmod p) - \left\lfloor \frac{p-1}{b} \right\rfloor$$

is an integer that depends only on $p \bmod b^2$.

Proof. Set $m = b^2$. For each $r \in \{1, \ldots, p-1\}$, define the slice index

$$n(r) = \left\lfloor \frac{mr}{p} \right\rfloor \in \{0, \ldots, m-1\}.$$

Since the slice width $p/m > 1$, each slice contains integers. On the slice with index n, both the digit function and the digit of the image under multiplication by b are constant:

$$\left\lfloor \frac{br}{p} \right\rfloor = \left\lfloor \frac{n}{b} \right\rfloor, \qquad \delta(br \bmod p) = n \bmod b.$$

Thus a slice contributes to the collision count exactly when

$$\left\lfloor \frac{n}{b} \right\rfloor = n \bmod b.$$

The diagonal slices are therefore

$$\begin{aligned} G &= \left\{ n \in \{0, \ldots, b^2{-}1\} : \lfloor n/b \rfloor = n \bmod b \right\} \\ &= \{ r(b{+}1) : 0 \leq r \leq b{-}1 \}. \end{aligned}$$

Hence

$$C(b \bmod p) = -1 + \sum_{n \in G} \left(\left\lfloor \frac{(n+1)p}{b^2} \right\rfloor - \left\lfloor \frac{np}{b^2} \right\rfloor \right),$$

where the -1 removes $r = 0$ from the $n = 0$ slice.

Write $p = b^2 Q_2 + a$ with $1 \leq a < b^2$ and $\gcd(a, b) = 1$. Then

$$\left\lfloor \frac{(n+1)p}{b^2} \right\rfloor - \left\lfloor \frac{np}{b^2} \right\rfloor = Q_2 + \left\lfloor \frac{(n+1)a}{b^2} \right\rfloor - \left\lfloor \frac{na}{b^2} \right\rfloor.$$

Since

$$\left\lfloor \frac{p-1}{b} \right\rfloor = bQ_2 + \left\lfloor \frac{a-1}{b} \right\rfloor,$$

it follows that

$$S(p) = -1 - \left\lfloor \frac{a-1}{b} \right\rfloor + \sum_{n \in G} \left(\left\lfloor \frac{(n+1)a}{b^2} \right\rfloor - \left\lfloor \frac{na}{b^2} \right\rfloor \right),$$

which depends only on $a = p \bmod b^2$. $\square$

Definition 2. The *collision periodic table* in base b is the function $\mathcal{T}\colon (\mathbb{Z}/b^2\mathbb{Z})^* \to \mathbb{Z}$ defined by $\mathcal{T}(a) = S(p)$ for any prime $p \equiv a \pmod{b^2}$ with $p > b^2$.

3. The Antisymmetry Law

Lemma 3 (Lag-1 closed form)**.** *Let $p > b^2$ be prime with $p \nmid b$, and write $p = b^2Q_2 + a$ with $1 \le a < b^2$ and $\gcd(a, b) = 1$. Then*

$$C(b, p) = bQ_2 + f(a) - 1,$$

where

$$f(a) = \#\{d \in \{0, \dots, b-1\} : (b+1)da \bmod b^2 \ge b^2 - a\}.$$

Proof. From Theorem 1,

$$C(b \bmod p) = -1 + \sum_{n \in G} \left(\left\lfloor \frac{(n+1)p}{b^2} \right\rfloor - \left\lfloor \frac{np}{b^2} \right\rfloor \right),$$

where $G = \{d(b+1) : 0 \le d \le b-1\}$. Writing $n = d(b+1)$ and $p = b^2Q_2 + a$ gives

$$\left\lfloor \frac{(n+1)p}{b^2} \right\rfloor - \left\lfloor \frac{np}{b^2} \right\rfloor = Q_2 + \left\lfloor \frac{(d(b+1)+1)a}{b^2} \right\rfloor - \left\lfloor \frac{d(b+1)a}{b^2} \right\rfloor.$$

Since $0 < a < b^2$, the final difference is either 0 or 1. It equals 1 exactly when the fractional part of $d(b+1)a/b^2$ is at least $1 - a/b^2$, equivalently when

$$(b+1)da \bmod b^2 \ge b^2 - a.$$

Thus each diagonal slice contributes Q_2 plus the indicator of that threshold event. Summing over $d = 0, \dots, b-1$ gives the formula. □

Theorem 4 (Antisymmetry)**.** *For every base $b \ge 3$ and every coprime class $a \bmod b^2$:*

$$\mathcal{T}(a) + \mathcal{T}(b^2 - a) = -1.$$

Proof. By Lemma 3, the collision count at multiplier $g = b$ satisfies $Q = \lfloor (p-1)/b \rfloor = bQ_2 + \lfloor (a-1)/b \rfloor$:

$$\mathcal{T}(a) = f(a) - 1 - \left\lfloor \frac{a-1}{b} \right\rfloor.$$

Three identities close the proof.

(i) Complementary counting. Write $r_d = (b+1)d \cdot a \bmod b^2$. For $d \ge 1$, $r_d \ne 0$ (since $\gcd(a, b^2) = 1$ and $\gcd(b+1, b^2) = 1$). The complement of r_d is $b^2 - r_d = (b+1)d \cdot (b^2 - a) \bmod b^2$. The condition $r_d \ge b^2 - a$ for $f(a)$ becomes $b^2 - r_d \le a$, which is the complement of the condition $b^2 - r_d \ge a$ for $f(b^2 - a)$. For $d \ge 1$: every d is counted by exactly one of

$f(a)$ or $f(b^2-a)$, except when $r_d = b^2 - a$ (counted by both). Including $d = 0$ (where $r_0 = 0$, counted by neither): $f(a) + f(b^2-a) = b$.

(ii) Floor sum. $(a-1) + (b^2-a-1) = b^2 - 2$. The remainders modulo b are $R = (a-1) \bmod b$ and $R' = (b^2-a-1) \bmod b = b-2-R$. Since $R + R' = b - 2 < b$: $\lfloor (a-1)/b \rfloor + \lfloor (b^2-a-1)/b \rfloor = (b^2 - 2 - (b-2))/b = b - 1$.

(iii) Assembly. $\mathcal{T}(a) + \mathcal{T}(b^2-a) = [f(a) + f(b^2-a)] - 2 - [\lfloor (a-1)/b \rfloor + \lfloor (b^2-a-1)/b \rfloor] = b - 2 - (b-1) = -1$. □

Corollary 5. *For every base $b \geq 3$:*

$$\mathcal{T}(b-1) = b - 2, \qquad \mathcal{T}(b^2 - b + 1) = -(b-1).$$

The range of $\mathcal{T}$ is contained in $[-(b-1),\ b-2]$.

Proof. For $a = b - 1$, the formula from the antisymmetry proof gives

$$\mathcal{T}(a) = f(a) - 1 - \left\lfloor \frac{a-1}{b} \right\rfloor = f(b-1) - 1.$$

Now

$$r_d = (b+1)d(b-1) \equiv -d \pmod{b^2},$$

so for each $d = 1, \ldots, b-1$ one has $r_d = b^2 - d \geq b^2 - b + 1 = b^2 - a$, while $d = 0$ is not counted. Thus $f(b-1) = b-1$, hence $\mathcal{T}(b-1) = b-2$.

The complementary class $b^2 - b + 1$ then satisfies

$$\mathcal{T}(b^2 - b + 1) = -1 - \mathcal{T}(b-1) = -(b-1)$$

by antisymmetry.

Finally, $d = 0$ is never counted in $f(a)$, so $0 \leq f(a) \leq b - 1$ for every coprime class a. Hence

$$\mathcal{T}(a) = f(a) - 1 - \left\lfloor \frac{a-1}{b} \right\rfloor \leq (b-1) - 1 = b - 2.$$

Applying antisymmetry gives $\mathcal{T}(a) \geq -(b-1)$. □

4. The Table

Base 10. The 40 coprime classes modulo 100 and their collision fingerprints at lag 1:

The table has 8 positive entries, 12 zeros, and 20 negative entries. The extremes are $\mathcal{T}(9) = +8$ and $\mathcal{T}(91) = -9$, confirming Corollary 5.

Universality across bases. The structural properties hold for every base tested:

	0	1	2	3	4	5	6	7	8	9
0	.	0	.	+2	.	.	.	0	.	+8
1	.	−1	.	−1	.	.	.	+1	.	−1
2	.	0	.	−2	.	.	.	+6	.	0
3	.	−1	.	−1	.	.	.	−3	.	−1
4	.	−4	.	0	.	.	.	−2	.	0
5	.	−1	.	+1	.	.	.	−1	.	+3
6	.	0	.	+2	.	.	.	0	.	0
7	.	−1	.	−7	.	.	.	+1	.	−1
8	.	0	.	−2	.	.	.	0	.	0
9	.	−9	.	−1	.	.	.	−3	.	−1

TABLE 1. Collision periodic table in base 10. Row is the tens digit, column is the units digit. Dots indicate classes not coprime to 100. The entry is $S = C(10 \bmod p) - \lfloor (p-1)/10 \rfloor$, constant for all primes $p > 100$ in that class.

b	$\varphi(b^2)$	$S_{\min}$	$S_{\max}$	+	0	−	Antisymmetry
3	6	−2	+1	1	2	3	$S(a) + S(-a) = -1$
5	20	−4	+3	4	6	10	$S(a) + S(-a) = -1$
6	12	−5	+4	2	4	6	$S(a) + S(-a) = -1$
7	42	−6	+5	10	11	21	$S(a) + S(-a) = -1$
8	32	−7	+6	7	9	16	$S(a) + S(-a) = -1$
10	40	−9	+8	8	12	20	$S(a) + S(-a) = -1$
12	48	−11	+10	11	13	24	$S(a) + S(-a) = -1$

TABLE 2. Summary across bases. In every case: $S_{\min} = -(b-1)$, $S_{\max} = b-2$, antisymmetry constant $= -1$, and negative classes outnumber positive.

5. The Sign Balance

Corollary 6 (Negative majority)**.** *Exactly $\varphi(b^2)/2$ classes have $\mathcal{T}(a) < 0$.*

Proof. The antisymmetry pairs each class a with $-a$, giving $\varphi(b^2)/2$ pairs. In each pair, $\mathcal{T}(a) + \mathcal{T}(-a) = -1 < 0$, so at least one member is negative. If $\mathcal{T}(a) = 0$ then $\mathcal{T}(-a) = -1 < 0$. If $\mathcal{T}(a) > 0$ then $\mathcal{T}(-a) = -1 - \mathcal{T}(a) < -1 < 0$. In every case, exactly one member per pair is negative. □

Corollary 7 (Mean)**.**

$$\frac{1}{\varphi(b^2)} \sum_{\substack{a \bmod b^2 \\ \gcd(a,b)=1}} \mathcal{T}(a) = -\frac{1}{2}.$$

Proof. Each pair $(a, -a)$ contributes $\mathcal{T}(a) + \mathcal{T}(-a) = -1$. There are $\varphi(b^2)/2$ pairs. The total is $-\varphi(b^2)/2$. □

Later Addition (2026)

This note was written to record the elementary lag-1 table and its antisymmetry. Later work reinterpreted the same table through Fourier analysis on $(\mathbb{Z}/b^2\mathbb{Z})^\times$. In that later framework, the table coefficients are supported on the primitive odd sector and factor through Bernoulli numbers and diagonal character sums. That analytic story belongs to the later transform and spectrum papers, not to the 2023 theorem package proved here.

For the purposes of the present note, the important point is only historical: the finite table came first. The Fourier and L-value interpretation was discovered later.

6. Remarks

Reading the table. The collision periodic table says: every prime carries a collision fingerprint determined by its residue class modulo b^2. In base 10, the prime 109 and the prime 50009 have the same fingerprint ($S = +8$) because both end in 09. The prime 191 and the prime 99991 share $S = -9$ because both end in 91. The structure is periodic, exact, and computable.

The -1. The antisymmetry constant is -1, not 0. The offset arises from the bilateral parity of the collision count (C is even) combined with the relationship between complementary bin structures. The complement map $r \mapsto p - r$ sends digit d to digit $b-1-d$, exchanging large and small bins. The -1 accounts for the parity mismatch between Q and Q' in paired classes.

The negative bias. Exactly half the classes are negative. This is a structural property of the antisymmetry: since $\mathcal{T}(a)+\mathcal{T}(-a) = -1 < 0$, every pair contains at least one negative member. The negative bias in the raw collision fluctuation is explained by this: the collision invariant at the specific multiplier b is systematically below the mean, and the periodic table shows this is not a statistical fluctuation but a structural certainty.

References

[1] A. S. Petty, *nfield: A structural analysis engine for fractional field invariants*, software. https://github.com/alexspetty/nfield

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